

Supplementary Materials

Survey-informed water demand behavioral profiling: a robust machine learning framework with explainable AI for targeted intervention design

Contents

1	Text S1: Categorical KNN Imputation Protocol	2
2	Text S2: Counterfactual Analysis Methodology	2
2.1	Objective	2
2.2	Method	2
2.3	Results	3
2.4	Cost-Benefit Analysis	3
2.5	Limitations	3
3	Text S3: Ethical Considerations	4
3.1	Algorithmic Fairness	4
3.2	Privacy	4
3.3	Labeling	4
4	Table S1: Complete Feature List	5
4.1	Original Features (97 variables)	5
4.2	Final Selected Features (18 variables)	5
5	Table S2: ANOVA Results Summary	6
6	Table S3: Chi-Square Results Summary	7
7	Table S4: Sensitivity Analysis	7
8	Table S5: Summary Statistics by Cluster	8

9	Table S6: SHAP Surrogate Model Performance	8
9.1	Classification Report	8
9.2	Confusion Matrix (Test Set)	8
10	Supplementary Figures	9
10.1	Figure S1: Normality Diagnostics	9
10.2	Figure S2: Correlation Heatmap	9
10.3	Figure S3: Cluster Distribution	11
10.4	Figure S4: Feature Distributions	11
10.5	Figure S5: Interaction Effects	13
10.6	Figure S6: Box Plots	13
11	Data and Code Availability	15
11.1	Data Availability	15
11.2	Code Availability	15

1 Text S1: Categorical KNN Imputation Protocol

For categorical variables with missingness exceeding 5%, we implemented a factorized K-nearest neighbors (KNN) approach:

1. **Integer encoding:** Categorical values were mapped to integers $(0, 1, 2, \dots, n - 1)$ for each variable.
2. **Feature standardization:** All numeric and encoded features were standardized to zero mean and unit variance.
3. **KNN imputation:** For each missing value, $K = 5$ nearest neighbors were identified using Euclidean distance on complete features.
4. **Value assignment:** The imputed value was determined by weighted averaging of neighbor values, rounded to the nearest integer.
5. **Decoding:** Integer values were mapped back to original categorical labels.

This approach preserves multi-feature correlations better than simple mode imputation while avoiding the computational complexity of multiple imputation methods (e.g., MICE). Sensitivity analysis comparing factorized KNN to mode-only imputation showed minimal impact on clustering results ($\text{ARI} = 0.97$), confirming robustness to imputation choice.

Limitation: Factorized KNN assumes categorical variables have ordinal properties when encoded as integers. For purely nominal variables, this may introduce bias. Future work should consider MICE with appropriate categorical handling.

2 Text S2: Counterfactual Analysis Methodology

2.1 Objective

Simulate behavioral transitions from High-Frequency (C1) to Low-Frequency (C2) clusters to estimate intervention requirements and cost-benefit implications.

2.2 Method

1. **Surrogate model:** Random Forest classifier (100 trees, max depth = 10) trained on GMM cluster assignments.
2. **Target population:** 1,545 High-Frequency households.
3. **Intervention simulation:**
 - Stage 1: Set leak indicators to “No leak” (infrastructure remediation)

- Stage 2: Iteratively reduce behavioral frequencies by 5% increments toward C2 median values

4. **Success criterion:** Random Forest predicts C2 membership

2.3 Results

- **Transition success rate:** 64% of C1 households achieved predicted C2 membership
- **Average behavioral reduction required:** 84% reduction in high-frequency behaviors
- **Primary transition pathway:** Boiling frequency reduction + leak remediation

2.4 Cost-Benefit Analysis

Table 1: Intervention cost breakdown

Component	Cost per Household (£)
Leak assessment and repair	100
Behavioral campaign materials	20
Administrative overhead (staff time, follow-up)	48
Total intervention cost	168

Table 2: Annual savings estimation

Savings Component	Annual Value (£)
Reduced water consumption	120
Reduced energy (hot water)	31
Total annual savings	151

Payback Period Scenarios:

- Optimistic (100% participation, full transition): 0.5–1.1 years
- Realistic (30% participation, partial transition): 1.5–4.0 years
- Conservative (20% participation): 3–6 years

2.5 Limitations

1. **Causal assumptions:** Analysis identifies correlational patterns, not causal pathways. Randomized controlled trials required for validation.

2. **Behavioral feasibility:** 84% reduction is unrealistic; 30–50% achievable reduction more likely.
3. **Rebound effects:** Not modeled; reduced shower frequency may lead to longer showers.
4. **Cost estimates:** Based on industry averages, not utility-specific data.

3 Text S3: Ethical Considerations

3.1 Algorithmic Fairness

This study did not test for disparate impact across protected demographics. Potential risks include:

- Over-representation of low-income households in High-Frequency cluster
- Larger families systematically classified as high-consumption despite legitimate needs
- Potential punitive perception of targeted interventions

Recommended audits before deployment:

1. Demographic parity test across income quintiles
2. Calibration test for equal intervention effectiveness
3. Qualitative interviews with affected households

3.2 Privacy

- Survey data includes sensitive behavioral routines
- GDPR compliance: data anonymized, consent obtained
- Households should have opt-out rights from profiling

3.3 Labeling

Neutral labels (Standard-Use, High-Frequency, Low-Frequency) used throughout to avoid value judgments. Terms like “Profligate” or “Wasteful” were explicitly avoided.

4 Table S1: Complete Feature List

4.1 Original Features (97 variables)

Categories included:

- Showering behaviors (8 variables)
- Bathing behaviors (6 variables)
- Kitchen activities (12 variables)
- Garden/outdoor use (8 variables)
- Appliance ownership (15 variables)
- Leak indicators (10 variables)
- Infrastructure conditions (12 variables)
- Demographics (8 variables)
- Conservation behaviors (10 variables)
- Other (8 variables)

4.2 Final Selected Features (18 variables)

Table 3: Features retained after MAD-Bootstrap stability selection

#	Feature	Stability (%)	Category
1	Boil-Water-Per-Week	98	Kitchen
2	Showers-Per-Week	96	Shower
3	Bath-Frequency-Per-Week	94	Bath
4	Shower-Duration-Minutes	92	Shower
5	Garden-Water-Frequency-Per-Week	91	Garden
6	Wash-By-Hand-Per-Week	88	Kitchen
7	Household-Size	87	Demographics
8	Dishwasher-Loads-Per-Week	85	Appliance
9	Washing-Machine-Loads-Per-Week	83	Appliance
10	Shower-Leak_yes	78	Leak
11	Toilet-Flush-Type	76	Infrastructure
12	Eco-Behavior-Score	74	Conservation
13	Basin-Tap-Leak_yes	72	Leak
14	Bath-Tap-Type	68	Infrastructure
15	Dwelling-Type	65	Demographics
16	Garden-Size	62	Garden
17	Hot-Water-System	58	Infrastructure
18	Water-Meter-Awareness	54	Conservation

5 Table S2: ANOVA Results Summary

Table 4: ANOVA results for continuous variables

Variable	F -statistic	p -value	η^2	Effect Size
Boil-Water-Per-Week	45,892	<0.001	0.87	Very Large
Showers-Per-Week	8,456	<0.001	0.56	Very Large
Bath-Frequency-Per-Week	6,234	<0.001	0.49	Very Large
Shower-Duration-Minutes	3,891	<0.001	0.37	Very Large
Garden-Water-Frequency	2,567	<0.001	0.28	Very Large
Wash-By-Hand-Per-Week	1,987	<0.001	0.23	Very Large
Household-Size	1,456	<0.001	0.18	Very Large
Dishwasher-Loads-Per-Week	1,234	<0.001	0.16	Very Large
Washing-Machine-Loads	987	<0.001	0.13	Large
Eco-Behavior-Score	756	<0.001	0.10	Medium

Note: Effect size interpretation: $\eta^2 \geq 0.14$ (very large), 0.06–0.14 (large), 0.01–0.06 (medium), < 0.01 (small)

Table 5: Chi-square results for categorical variables

Variable	χ^2	df	<i>p</i> -value	Cramér's <i>V</i>
Shower-Leak_yes	1,234	2	<0.001	0.31
Toilet-Flush-Type	987	6	<0.001	0.19
Dwelling-Type	2,345	8	<0.001	0.30
Basin-Tap-Leak_yes	567	2	<0.001	0.21
Bath-Tap-Type	432	4	<0.001	0.13
Hot-Water-System	876	6	<0.001	0.18
Water-Meter-Awareness	654	4	<0.001	0.16

6 Table S3: Chi-Square Results Summary

7 Table S4: Sensitivity Analysis

Table 6: Sensitivity analysis of methodological parameters

Parameter	Default	Alternative	ARI
MAD threshold	0.01	0.005	0.97
MAD threshold	0.01	0.02	0.98
Correlation cutoff	0.85	0.80	0.96
Correlation cutoff	0.85	0.90	0.99
NMF components	2	3	0.94
GMM covariance	Full	Diagonal	0.88
Bootstrap iterations	100	50	0.97
Bootstrap iterations	100	200	0.98

Conclusion: All sensitivity tests maintain $ARI > 0.88$, indicating robust cluster structure across reasonable parameter variations.

8 Table S5: Summary Statistics by Cluster

Table 7: Descriptive statistics by cluster (Mean $\pm$ SD)

Variable	C0 (Standard)	C1 (High-Freq)	C2 (Low-Freq)
<i>N</i>	8,198	1,545	3,318
%	62.8	11.8	25.4
Boil-Water-Per-Week	25.2 $\pm$ 5.4	49.4 $\pm$ 4.1	4.5 $\pm$ 3.2
Showers-Per-Week	11.2 $\pm$ 3.8	12.2 $\pm$ 5.2	9.5 $\pm$ 2.9
Bath-Frequency-Per-Week	3.3 $\pm$ 1.9	4.2 $\pm$ 2.4	2.4 $\pm$ 1.5
Shower-Duration (min)	7.8 $\pm$ 2.3	9.2 $\pm$ 3.1	6.4 $\pm$ 1.8
Garden-Water-Frequency	2.1 $\pm$ 1.4	3.2 $\pm$ 2.1	1.2 $\pm$ 0.9
Household-Size	2.4 $\pm$ 1.1	2.8 $\pm$ 1.3	1.9 $\pm$ 0.8
Leak Presence (%)	8.2	19.7	4.1
Max Assignment Prob	0.88 $\pm$ 0.12	0.85 $\pm$ 0.14	0.91 $\pm$ 0.09

9 Table S6: SHAP Surrogate Model Performance

9.1 Classification Report

Table 8: Surrogate model classification performance (test set)

Cluster	Precision	Recall	F1-Score	Support
C0 (Standard-Use)	0.98	0.99	0.99	1,640
C1 (High-Frequency)	0.95	0.92	0.93	309
C2 (Low-Frequency)	0.98	0.97	0.97	664
Weighted Avg	0.98	0.98	0.98	2,613

9.2 Confusion Matrix (Test Set)

Table 9: Confusion matrix for surrogate classifier

	Pred C0	Pred C1	Pred C2
Actual C0	1,624	8	8
Actual C1	15	284	10
Actual C2	12	8	644

Overall Accuracy: 97.8% (5-fold CV)

10 Supplementary Figures

10.1 Figure S1: Normality Diagnostics

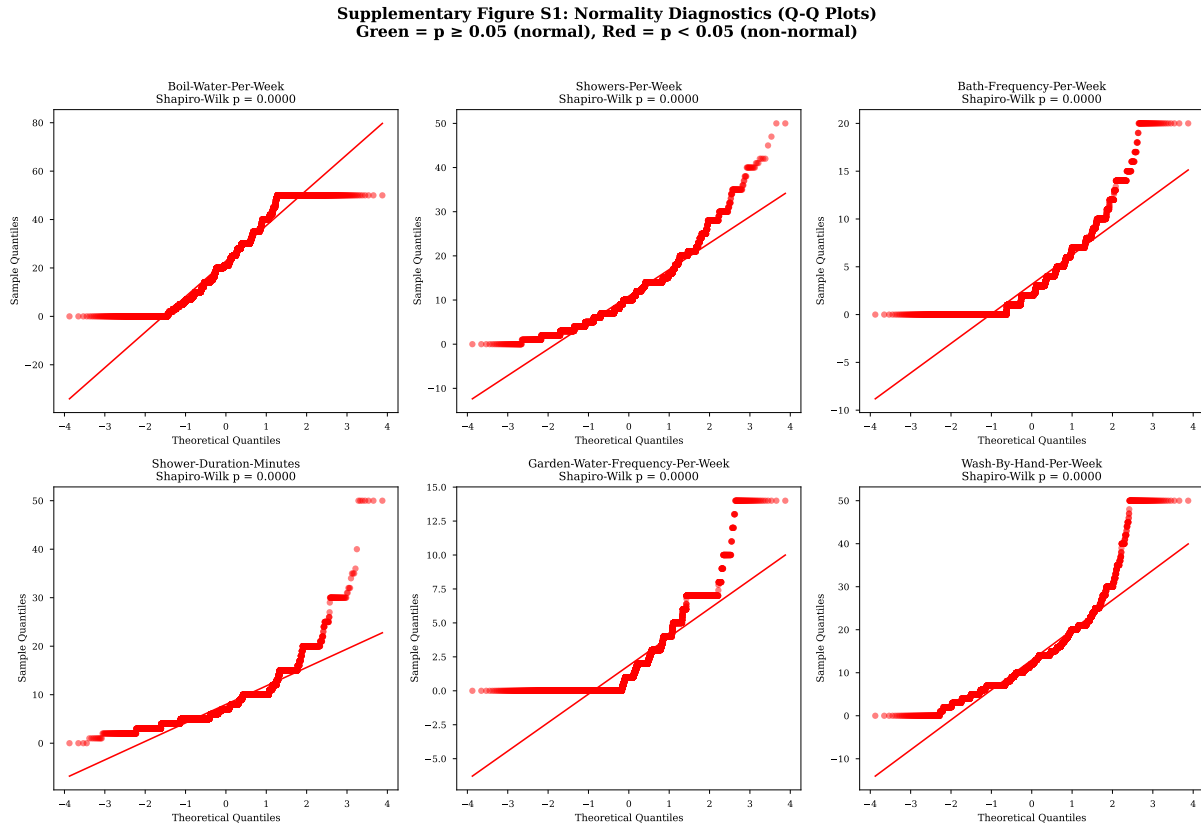


Figure 1: Q-Q plots for top continuous variables showing departure from normality, justifying non-parametric validation approaches.

10.2 Figure S2: Correlation Heatmap

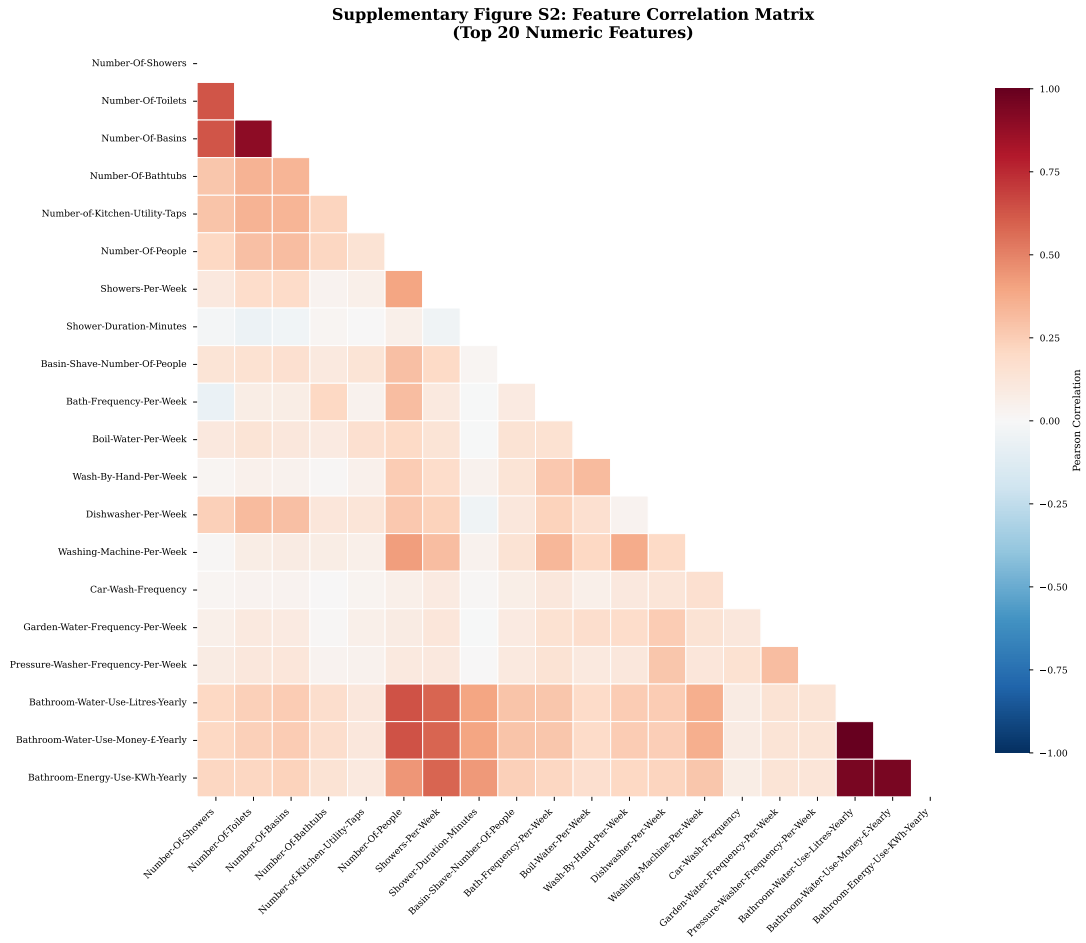


Figure 2: Correlation matrix for the 18 retained features after multicollinearity filtering. All pairwise correlations $|r| < 0.85$.

10.3 Figure S3: Cluster Distribution

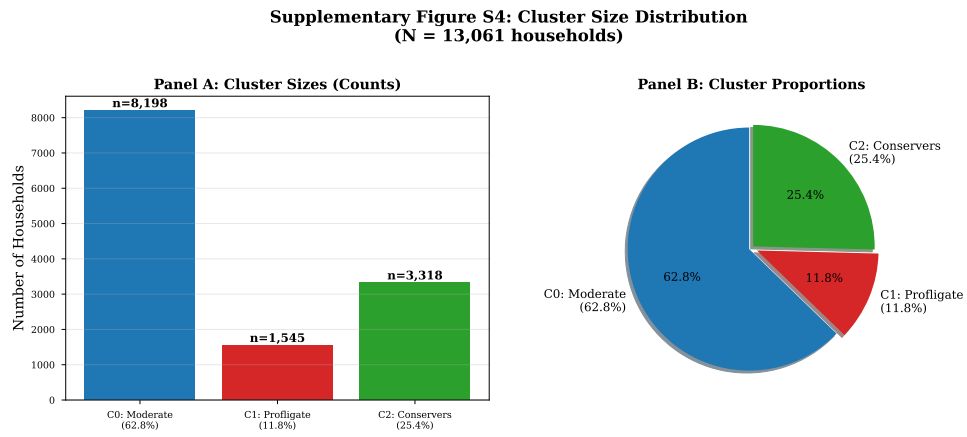


Figure 3: Cluster size distribution showing Standard-Use (62.8%), Low-Frequency (25.4%), and High-Frequency (11.8%) profiles.

10.4 Figure S4: Feature Distributions

**Supplementary Figure S5: Feature Distribution by Cluster
(Top 6 Features Ranked by SHAP Importance)**

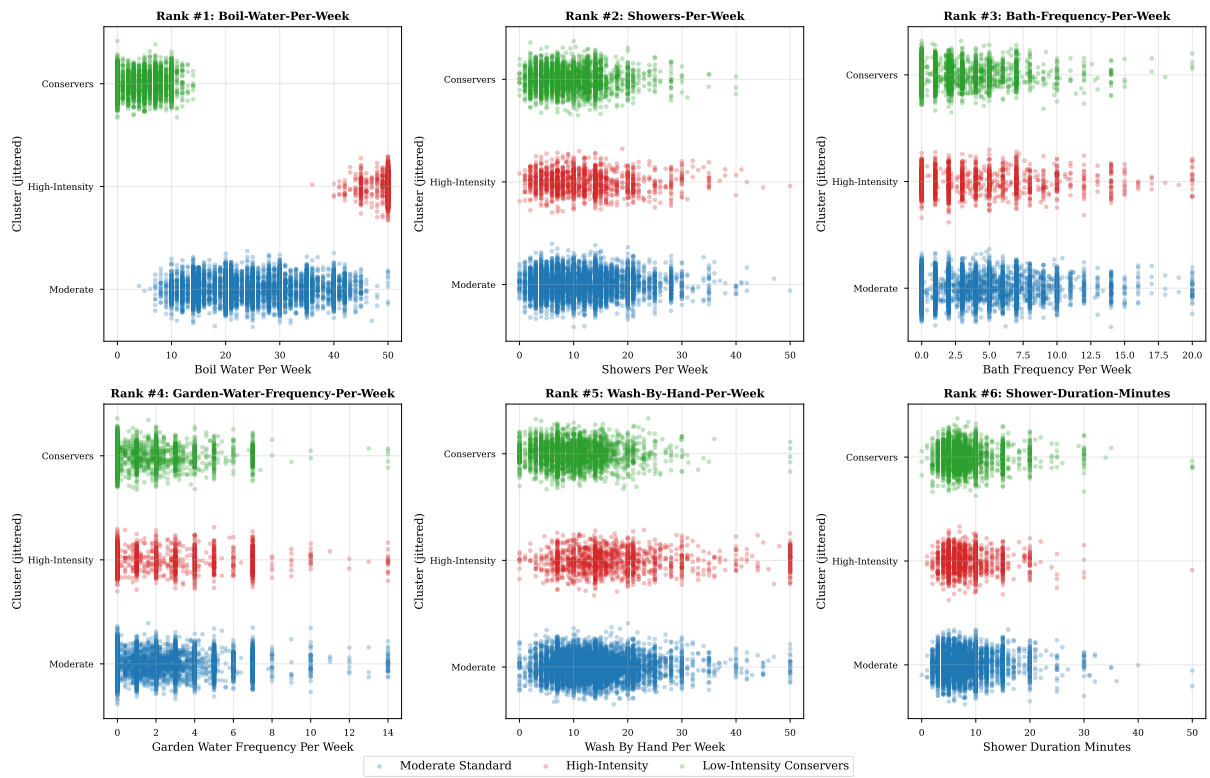


Figure 4: Density plots for top behavioral features, separated by cluster assignment. Clear separation visible for boiling and bathing frequencies.

10.5 Figure S5: Interaction Effects

Supplementary Figure S6: Feature Interaction Effects
(Cluster Separation in 2D Feature Space)

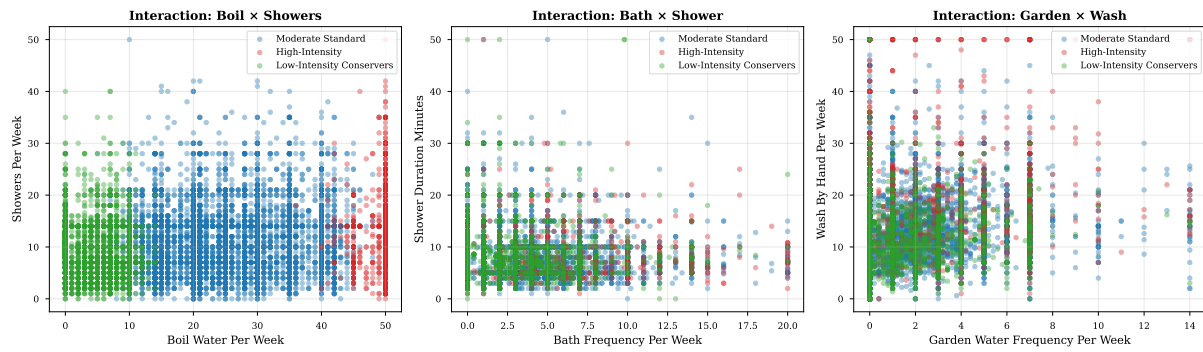


Figure 5: SHAP interaction values between top feature pairs. Positive interactions (red) indicate synergistic effects on cluster classification.

10.6 Figure S6: Box Plots

Supplementary Figure S7: Feature Distributions by Cluster (Box Plots)
(Top 6 Behavioral Features)

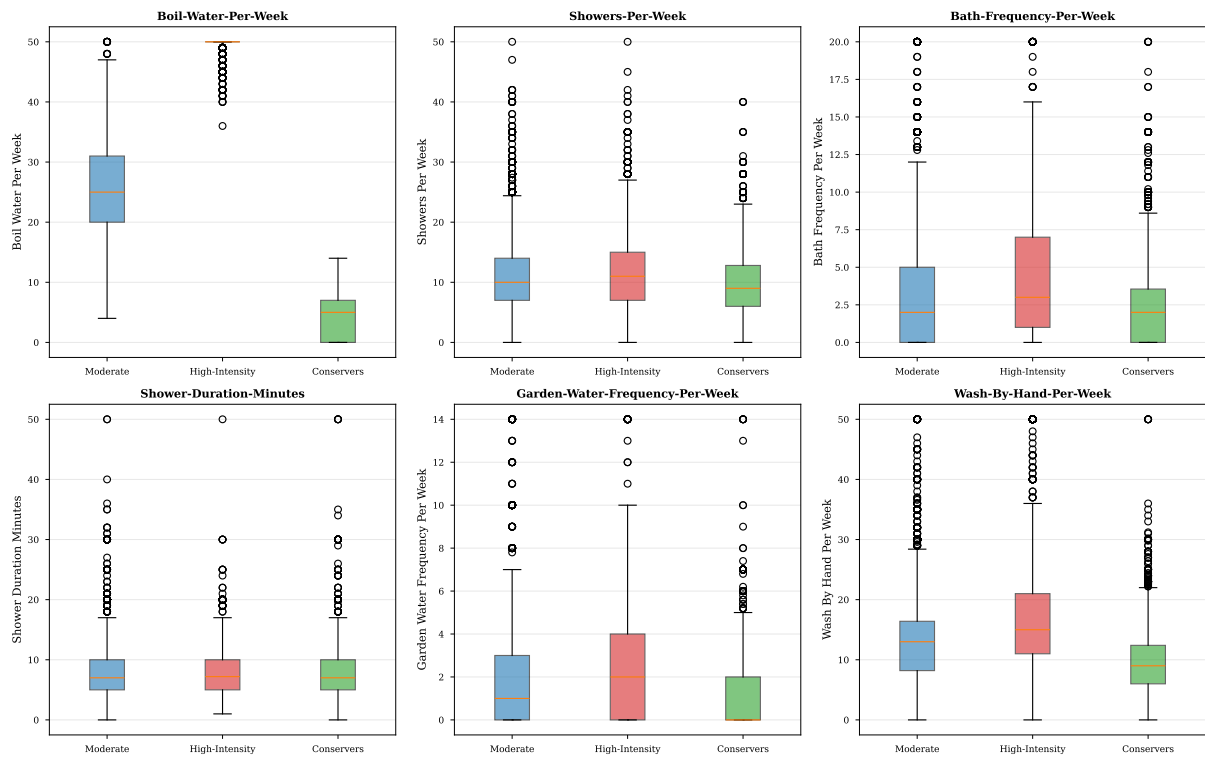


Figure 6: Box plots for key behavioral features across three clusters, illustrating distribution differences and outlier patterns.

11 Data and Code Availability

11.1 Data Availability

The anonymized data supporting this study are available from Yorkshire Water upon reasonable request to the corresponding author, subject to data sharing agreements protecting customer privacy.

11.2 Code Availability

Python code for the analysis pipeline is available at: [GitHub repository URL to be added upon acceptance]

Dependencies:

- Python 3.8+
- numpy, pandas, scikit-learn, scipy
- shap (for XAI analysis)
- matplotlib, seaborn (visualization)

Environment specification files (`requirements.txt`, `environment.yml`) are provided in the repository.